Descriptive Statistics

A Quick-Reference Study Guide — Session 1

What's inside

1. What is Statistics?

Statistics is a branch of mathematics involving the collection, analysis, interpretation, and presentation of data. It gives us tools to make sense of large amounts of information and to draw conclusions or make decisions based on it.

Where it's used:

Field	Example Use
Business	Customer behavior analysis, demand forecasting
Medicine	Clinical trials (drug efficacy), epidemiology (risk factors)
Government & Politics	Surveys, polling
Environmental Science	Climate research

2. Types of Statistics

Descriptive Statistics	Inferential Statistics
Summarizes and describes the main features of a dataset. No predictions — just describes what the data shows.	Uses a sample to make predictions/conclusions about a larger population, using

*Note: this guide focuses on **Descriptive Statistics** — Section 4 previews inferential topics for context.*

3. Population vs. Sample

- **Population** — the entire group of individuals/objects being studied (e.g., all students in a school).
- **Sample** — a smaller subset selected from the population, used to estimate population characteristics (e.g., 100 randomly chosen students).
- **Parameter** — a numerical characteristic of the *population* (usually unknown).
- **Statistic** — a numerical characteristic of the *sample* (used to estimate the parameter).

Examples

- All cricket fans (population) vs. fans present in the stadium (sample)
- All students (population) vs. students who visit college for lectures (sample)

Things to be careful about when creating samples:

- Sample size
- Randomness
- Representativeness

4. Inferential Statistics — Topics Preview

Inferential statistics uses a sample to draw conclusions about a population. Key topics include:

Topic	What it does
Hypothesis Testing	Tests a claim about a population parameter using sample data
Confidence Intervals	Estimates a range of values a population parameter could take
ANOVA	Compares means across multiple groups for significant differences
Regression Analysis	Models relationships between a dependent and independent variable(s)
Chi-Square Tests	Tests independence/association between two categorical variables
Sampling Techniques	Ensures the sample represents the population (e.g., random sampling)
Bayesian Statistics	Updates the probability of an event as new evidence arrives

5. Types of Data

Category	Subtype	Description	Example
Categorical (Qualitative)	Nominal	No inherent order	Gender: Male / Female
	Ordinal	Has a meaningful order	Rating: Bad / Average / Good
Numerical (Quantitative)	Discrete	Countable values	Age, Rank
	Continuous	Measurable, can take any value	Weight

6. Measures of Central Tendency

A single value that represents the 'typical' or central value of a dataset.

Mean

Sum of all values divided by the number of values.

Population Mean: μ = (Σ x■) / N | Sample Mean: x■ = (Σ x■) / n

N = number of items in the population | *n = number of items in the sample*

Median

The middle value in the dataset once it's arranged in order.

Mode

The value that appears most frequently in the dataset.

Weighted Mean

Sum of the products of each value and its weight, divided by the sum of the weights. Used when values in the dataset carry different importance or frequency.

Trimmed Mean

Calculated by removing a set percentage of the smallest and largest values, then taking the mean of what remains. The removed percentage is called the trimming percentage. This reduces the influence of outliers.

7. Measures of Dispersion

Describes the spread or variability of a dataset — how data is distributed around the central tendency (mean, median, or mode).

Range

The difference between the maximum and minimum values. Simple, but sensitive to outliers.

Variance

The average of the squared differences between each data point and the mean. Measures average distance from the mean; useful for comparing dispersion across datasets with different means.

Population Variance: $\sigma^2 = \Sigma(x - \mu)^2 / N$

Sample Variance: $s^2 = \Sigma(x - x\blacksquare)^2 / (n - 1)$

Why n – 1 for samples? It corrects for bias, giving a more accurate estimate of the population variance (this is called Bessel's correction).

Standard Deviation

The square root of the variance. Widely used because it's expressed in the same units as the original data, making it easier to interpret.

Population SD: $\sigma = \sqrt{\Sigma(x - \mu)^2 / N}$ | **Sample SD:** $s = \sqrt{\Sigma(x - x\blacksquare)^2 / (n - 1)}$

Mean Absolute Deviation (MAD)

The average of the absolute distances between each data point and the mean.

$MAD = \Sigma \mid x\blacksquare - x\blacksquare \mid / n$

Coefficient of Variation (CV)

The ratio of the standard deviation to the mean, expressed as a percentage. It's dimensionless, which makes it useful for comparing variability between datasets with different means or units (common in biology, chemistry, and engineering).

$CV = (\text{Standard Deviation} / \text{Mean}) \times 100\%$

8. Visualizing Data

Univariate Analysis (One Variable)

Categorical data: Frequency Distribution Table, Bar Chart, Pie Chart

Term	Meaning
Frequency	Number of times each value/category occurs
Relative Frequency	Frequency ÷ total number of observations
Cumulative Frequency	Running total of frequencies up to and including the current category

Numerical data: Frequency Distribution Table (with bins/buckets) + Histogram

Histogram Shapes

Shape	Description
Symmetric	Data is evenly distributed around the center (bell-shaped)
Bimodal	Two distinct peaks in the distribution
Left-Skewed	Long tail on the left side (low outliers pull the tail left)
Right-Skewed	Long tail on the right side (high outliers pull the tail right)
Uniform	All bars roughly equal height — no clear peak
No Pattern	Irregular, random-looking distribution

Bivariate Analysis (Two Variables)

Variable Types	Best Chart
Categorical – Categorical	Contingency Table / Crosstab
Numerical – Numerical	Scatter Plot
Categorical – Numerical	Box Plot (grouped comparisons)

End of guide — compiled from Session 1 on Descriptive Statistics.